ASSIGNMENT 5

Unit Covered	Unit V — Motion Tracking, Object Recognition & Shape Representation
Suggested Tools	Python (OpenCV, NumPy), a short video with a moving object

1. Source or record a video where a single object moves across the frame and is briefly, partially occluded at least once (walks behind a pole, pillar, or another object).
2. Detect the object in the first frame (manual bounding box is fine) and implement a Kalman filter to predict and update its position across subsequent frames. Overlay the predicted vs. measured position on each frame.
3. Specifically show and explain what the Kalman filter predicts during the occluded frames (where no measurement is available) and how well the object is "re-acquired" once it reappears.
4. Briefly discuss, referencing observability, why a fast, erratically-moving object would be harder for your filter to track reliably than a slow, smooth one.
5. For the recognition half: collect 4–5 reference images each of two or three distinct objects/people (with permission) or use a small public image set. Implement an appearance-based recognition approach using image eigenspaces (Eigenfaces/PCA) — or an alignment-based matching approach if you use non-face objects — and test it on a held-out image of each class.
6. Report your recognition accuracy on the held-out set and discuss at least two conditions (pose, lighting, occlusion, scale) under which your recognizer's accuracy drops, with example failure images.